

The background is a soft, watercolor-style illustration in shades of light green and white. It features various types of green leaves and foliage, including large monstera leaves with characteristic splits and smaller, rounded leaves, scattered around the edges of the frame. The overall aesthetic is clean, fresh, and nature-inspired.

PLANTY

Revolutionizing Crop Disease Identification

Empowering Farmers with Technology

By Masterminds

THE REALITY OF CROP DISEASES

- Farmers lose billions annually due to undiagnosed diseases.
- Delayed identification leads to reduced yields and financial loss.
- Need for quick, accurate solutions for this problem.



EMBRACING TECHNOLOGY IN FARMING

- Technology can transform agricultural practices.
- Mobile apps offer real-time information and support.
- Empower farmers to make informed decisions for better productivity.



MEET PLANTY

- An innovative app designed for crop disease identification.
- User-friendly interface: upload photos for instant diagnosis.
- Tailored recommendations for treatment and prevention

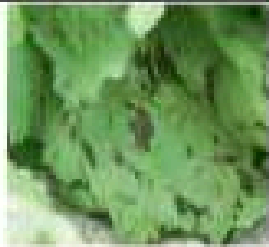
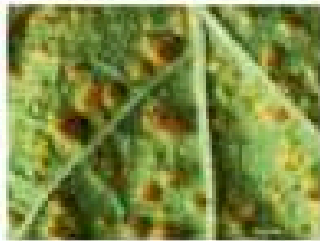
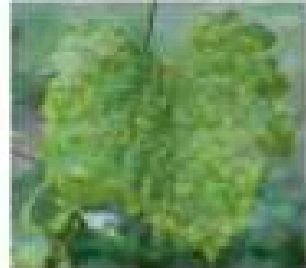
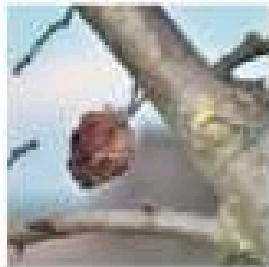
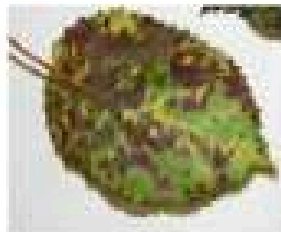
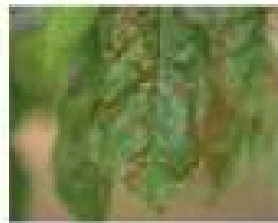
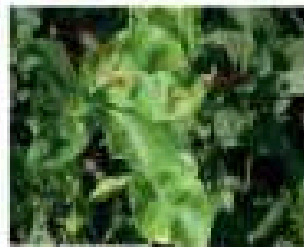
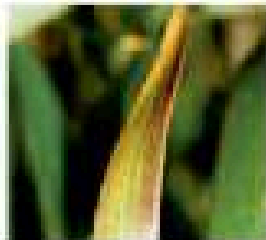


VISION FOR FUTURE OF FARMING

- Make every farmer equipped with the tools to safeguard their crops.
- Enhancing productivity and promoting sustainable practices.
- Planty aims to build a resilient agricultural landscape for all.



CROP DISEASES

Bacterial Diseases	 Softspot	 Spot	 Wilt
	 Molds	 Rust	 Mildew
Fungal Diseases	 Rots	 Spots	 Wilts
	 Mottling	 Distortion	 Dwarfing

IMPACT OF CROP DISEASES

- Nearly 40% of global food production is lost to diseases and pests.
- Farmers often lack timely information to address these challenges.
- Planty bridges the gap for proactive disease management.



A SURVEY



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A Survey on Plant Leaf Disease Detection

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ABSTRACT

Deep learning constitutes a recent, modern technique for image processing with accurate results. Many techniques of deep learning and image processing are used for leaf disease detection and classification. Deep learning techniques such as CNN, Fast RCNN, Faster RCNN, and Mask RCNN, and image processing techniques such as image preprocessing, segmentation, feature extraction etc. are used for disease detection. As per the survey, deep learning technique provides high accuracy than image processing technique. Plant leaf disease detection has wide range of applications available in various fields such as Biological Research and in Agriculture Institute. Agricultural productivity is something on which economy highly depends. This paper provides an overview of various techniques that are used for Plant Leaf Disease Detection. It also covers survey on different diseases classification techniques that can be used for plant leaf disease detection. Some authors are describing to find leaf diseases using various methods and to recommend the various implementations.

KEYWORDS: Leaf Disease Detection, Deep Learning, Image Processing, Feature Extraction, Convolutional Neural Network

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WHY CHOOSE PLANTY?

- Quick and easy disease identification.
- Informed decision-making for better crop management.
- Increased yield and reduced losses.



SIMPLE & EFFECTIVE

- Upload a photo of the affected plant.
- ML model analyzes the image and identifies potential diseases.
- Receive instant feedback and treatment recommendations.



BUILDING A FARMER COMMUNITY

- Connect farmers from diverse backgrounds to share knowledge and experiences.
- Discuss issues, solutions, and best practices in real-time.
- Foster collaboration and support among farmers.



INTELLIGENT ASSISTANCE AT YOUR FINGERTIPS



- Integrated Gen AI model to provide instant answers to farmers' questions.
- Access expert knowledge and practical advice tailored to specific issues.
- Reduces time spent searching for solutions, enhancing productivity.



WHY PLANTY STANDS OUT

- Combines disease identification with community support and AI assistance.
- Empowers farmers with knowledge and resources to make informed decisions.
- Creates a collaborative ecosystem that benefits all users.



SUGGETIONS

- Start by focusing on a single crop, then gradually expand.
- Is it feasible for Sri Lankan farmers to adopt this kind of technology?



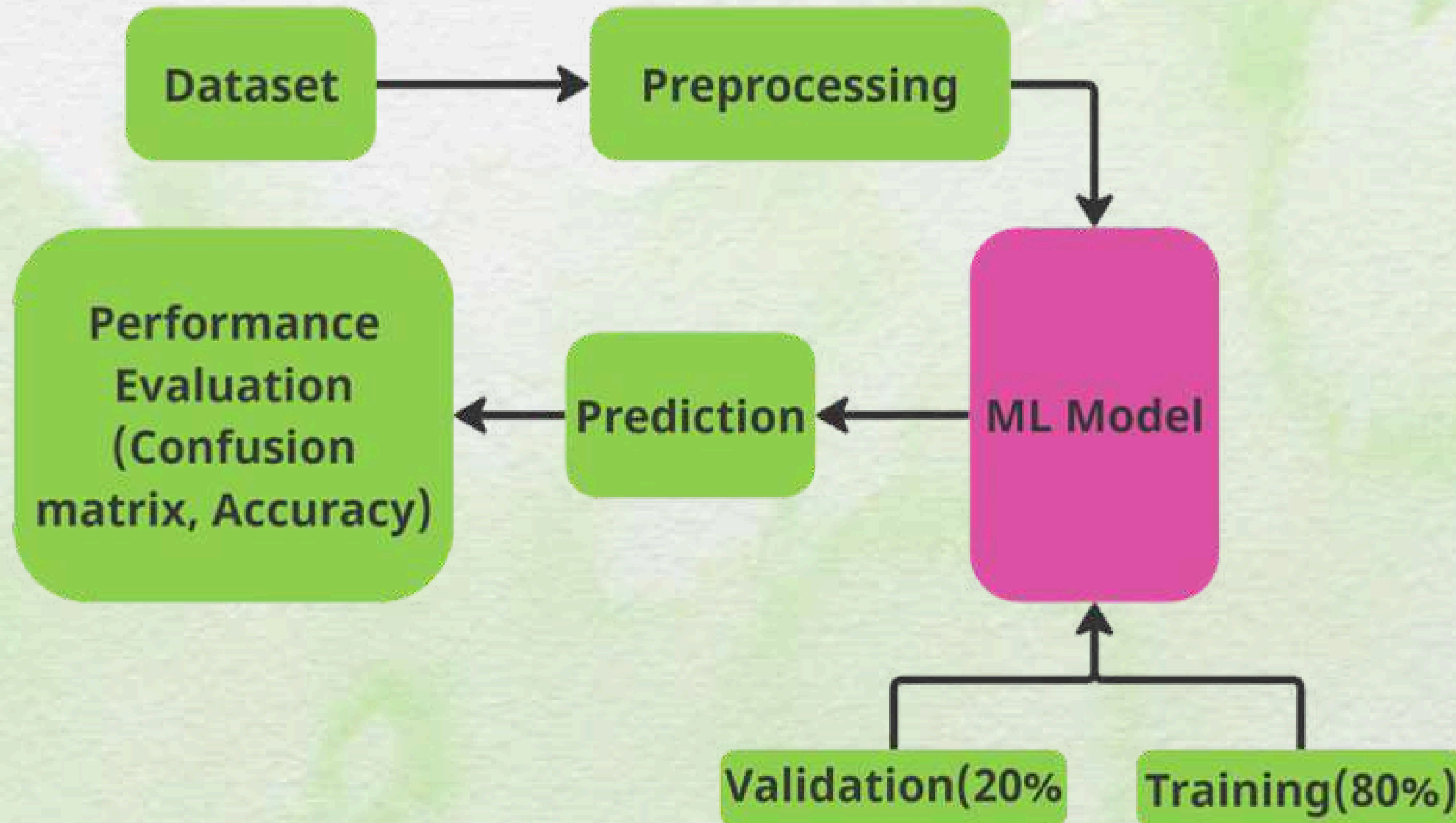
ML MODEL

Plant Diseases Dataset



- Dataset includes around 87,000 RGB images of healthy and diseased crop leaves, categorized into 38 classes.
- Divided into 80 : 20 training and validation sets, with an additional 33 test images for predictions.

WORK FLOW DIAGRAM



SUSTAINABLE REVENUE STREAMS

- **Ad Revenue:** Targeted ads relevant to farmers (seeds, tools, fertilizers).
- **Sponsored Content:** Partner with agricultural brands for sponsored posts and promotions.
- **Premium Model:** Basic features are free; premium features available through subscription.



CAPITALIZING ON A GROWING MARKET

- Global agriculture technology market projected to grow significantly.
- Demand for digital solutions in agriculture is on the rise.
- Increasing smartphone penetration among farmers opens new opportunities.



REACHING OUR AUDIENCE

- **Social Media Campaigns:** Engage farmers on Facebook, WhatsApp
- **Partnerships:** Collaborate with agricultural organizations and cooperatives.
- **Webinars and Workshops:** Educate farmers about app features & benefits.



MEASURING OUR IMPACT

- **User engagement:** Track community interactions and question responses.
- **Revenue growth:** Monitor ad revenue and premium subscriptions.
- **User satisfaction:** Gather feedback to continuously improve features.



VISION FOR GROWTH

- Expand the community features to include regional groups.
- Introduce more advanced AI capabilities for personalized recommendations.
- Explore partnerships with research institutions for cutting-edge agricultural insights.



SUSTAINABLE DEVELOPMENT GOALS



- Enhances food security by empowering farmers to identify and manage crop diseases effectively.
- Supports increased agricultural productivity and sustainable farming practices.

SUSTAINABLE DEVELOPMENT GOALS



- Encourages adaptive farming techniques that enhance resilience to climate impacts.
- Facilitates sharing of knowledge on sustainable practices within the farming community.

SUSTAINABLE DEVELOPMENT GOALS



- Promotes sustainable agricultural practices through better crop management.
- Reduces waste by preventing crop losses due to diseases.

The background is a soft, watercolor-style illustration. It features various green leaves and foliage. In the top left, there are large, dark green monstera leaves with characteristic splits. In the top right, there are smaller, rounded green leaves on a thin branch, set against a light blue and green wash. In the bottom left, more rounded green leaves are visible. In the bottom right, there are more monstera leaves. The overall color palette is a mix of light greens, teal, and off-white, creating a fresh and natural feel.

Q & A